

DeepMI: Deep Learning for Multiclass Motor Imagery Classification

Waseem Abbas¹ and Nadeem Ahmad Khan²

Abstract—In Brain-Computer Interface (BCI) Research, Electroencephalography (EEG) has obtained great attention for biomedical applications. In BCI system, feature representation and classification are important tasks as the accuracy of classification highly depends on these stages. In this paper, we propose a model in which Common Spatial Pattern (CSP) is used to discriminate inter-class data using co-variance maximization and Fast Fourier Transform Energy Map (FFTEM) is used for feature selection and mapping of 1D data into 2D data (energy maps). Convolutional Neural Network is used for classification of multi-class Motor Imagery (MI) signals. Further, this paper investigates near-optimal parameter selection for feature mapping, frequency bands selection, and temporal segmentation. It is shown that our proposed method outperformed the reported methods by achieving 0.61 mean kappa value.

I. INTRODUCTION

A few people experience the ill effects of neurological malady and face challenges in strolling, composing and talking due to dishonorable engine control. Such sorts of ailments have an effect on their correspondence and they are considered as a burden on society [1]. By analyzing patterns of brain signals of such handicapped persons, a command can be generated for prosthetic limbs or other devices for assistive control. EEG can be measured in invasive or non-invasive manners where the non-invasive method of EEG measurement has obtained more attention because of its safety and simple installations [2]. Motor Imagery (MI) is connected to the tangible engine cortex range of human cerebrum which can be recorded by focal, parietal and worldly anodes of the global 10 20 cathode position framework.

Advancement in neural networks and its applications in such complicated tasks brought a perspective approach to MI classification in the branch of machine learning. Convolutional Neural Network (CNN) is a novel classification approach based on images fed as input. CNN can be beneficial in MI classification as it learns features automatically as compared to classical (signal processing) approaches which utilize handcrafted features obtained using signal and image processing techniques.

In last the decade, a number of publications have been reported in an area of multi-class MI classification using CNN. In [3], researchers used Steady State Visually Evoked Potential (SSVEP) for classification of EEG signal. On a subject, visual stimulus of specific frequency was used where CNN of four layers with Fourier Transform (FT) in the

second layer was tried out. They were able to achieve 97% classification accuracy. They argued that switching to frequency domain from time domain could increase classification accuracy.

In [4], researchers described different applications of SSEVP with CNN. They used four layer CNN where a hidden layer of L2 Fast Fourier Transform (FFT) was used for extraction of frequencies to analyze signals in frequency domain. They generated 2D images of filtered EEG signals with the size of 4×512 which were fed to the CNN classifier. They obtained 80% accuracy using this network with 1000 epochs. CNN was used for P300 in [5] where researchers obtained 90% accuracy. They analyzed both features: time domain and frequency domain, separately. They generated the 2D image of size 64×64 from EEG data obtained from 64 channels. They used several different architectures of CNN and kernel in vector form was used as compared to kernels in matrix form to prevent mixing of time and space features. They used 8 most prominent channels for analysis using weight layer. Another paper in MI classification problem, [9] describes an analysis on the classification of multi-class MI signals. They used band power features and Auto-encoder for feature reduction with LDA classifier in One Versus Rest (OVR) mode and obtained 0.56 mean kappa value. Band power could be a promising feature for the classification of multi-class motor imagery data. In this paper, we proposed a novel approach for multi-class EEG data classification using Fast Fourier Transform Energy Map (FFTEM) as the final feature set with Convolutional Neural Network (CNN) as the classifier.

II. PROPOSED METHOD

Proposed model consist of four stages: Preprocessing, feature extraction, feature selection, and classification. Structure of proposed model is shown in Fig. 1. We used Common Spatial Pattern (CSP) and Fast Fourier Transform Energy Maps (FFTEM) for feature computation and selection whereas a Convolutional Neural Network (CNN) is proposed as a classifier with the novel features.

A. Preprocessing

To maximize classification accuracy, artifacts present in raw EEG data must be filtered out and selection of optimal frequency bands can help in discrimination of the classes present in MI data. In order to process only optimal frequency bands, we extracted μ (8-13 Hz) and β (13-30 Hz) bands using 5th order Butter-worth filter as these bands are found most effective in literature [7]. Temporal segmentation is another important factor in EEG signal processing as

¹Waseem Abbas is with Department of Computer Sciences, Lahore University of Management Sciences, opposite Sector U, D.H.A., Lahore, Pakistan waseem.abbas@lums.edu.pk

²Nadeem Ahmad Khan is with Faculty of Electrical Engineering, Lahore University of Management Sciences, opposite Sector U, D.H.A., Lahore, Pakistan n.khan@lums.edu.pk

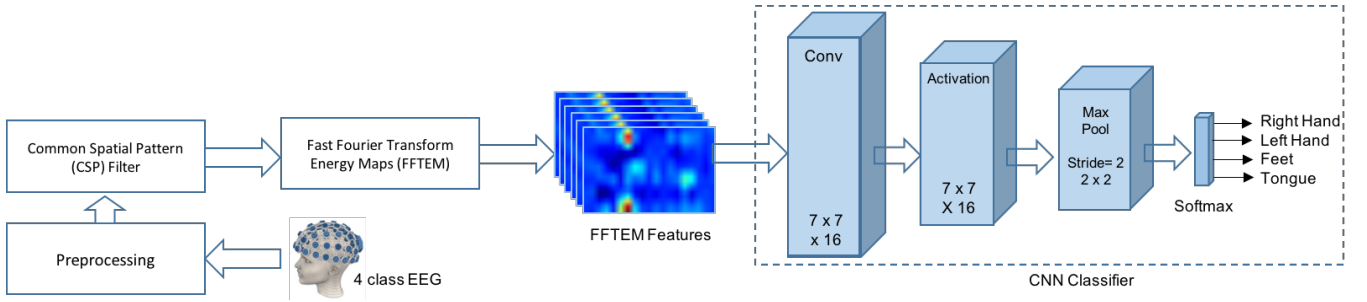


Fig. 1. Proposed Model, Input: 4 class motor imagery data, output: class confidence, Key terms: FFTEM: Fast Fourier Transform Energy Map

Dataset contains start and stop queues at the start and the end of each trial. We selected average time segment of 2.5-5.0 sec over all subjects to drop redundant temporal information. These temporal windows are selected on the basis of experimentation where we used small chunks of time segments to analyze classification performance in a norm. These temporal windows can be visualized in Fig. 2.

B. Common Spatial Pattern (CSP)

The principle of CSP algorithm is based on the selection of spatial filters from multi-channel signals for given recordings of two class distributions and maximizes variance for a class and minimizes variance for other class [6]. This can be achieved by determining spatial patterns via simultaneous diagonalization of covariance matrices for all classes. Covariance can be maximized using Raleigh criterion as:

$$\mathbf{R}(\mathbf{w}) = \frac{\mathbf{W}^T \sum_1 \mathbf{W}}{\mathbf{W}^T (\sum_1 + \sum_2) \mathbf{W}} \quad (1)$$

where $\sum_1$ and $\sum_2$ are covariance matrices of class 1 and class 2 and $\mathbf{W}$ is data matrix. This can be solved using following optimization problem:

$$\max_{\{w\}} \mathbf{W}^T \sum_1 \mathbf{W} \quad (2)$$

which satisfies the following relation:

$$\mathbf{W}^T (\sum_1 + \sum_2) \mathbf{W} - \mathbf{I} = 0 \quad (3)$$

Transformation matrix $\mathbf{Y}$ is obtained using:

$$\mathbf{Y} = \mathbf{W} \mathbf{X} \quad (4)$$

where $\mathbf{X}$ is input vector of a single trial EEG.

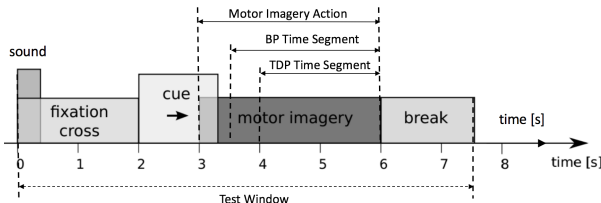


Fig. 2. Paradigm for extraction of a single trial

C. Fast Fourier Transform Energy Map (FFTEM)

Fast Fourier Transform Energy Map (FFTEM) has been used as feature selection method which exploits both the spectral and energy information hidden in EEG data. Further, FFTEM is less computational expensive as compared to wavelet and FFT which turns FFTEM as a strong candidate for 2D (energy vs frequency) feature mapping. FFTEM is applied to the discriminative features obtained from CSP filters to compute energy maps correspond to frequency bands and EEG channels. These energy maps are used to train the CNN classifier. In FFTEM, each channel of EEG data (transformed in the spatial domain) is mapped in the frequency domain using the following equation:

$$\mathbf{H}_i = |\mathbf{FFT}(\mathbf{Y}_i)|, i = \overline{1, N} \quad (5)$$

where $\mathbf{H}$ is the 2D energy map between channels and frequency spectrum and $\mathbf{Y}$ is the output of CSP block. An example of energy map (for entire frequency band) is shown in Fig. 3.

D. Convolutional Neural Networks (CNN) Classifier

Convolutional Neural Networks work on the principle of multi-layer perceptron. CNN has cured attention of researchers because of its ability to work with 2D dimensional data. CNN consist of one or multiple convolutional layers and multiple layers form a chain of non-linear filters. We used a baseline CNN model to classify multi-class data which can be seen in Fig. 1. CNN classifier consists of convolutional layer, normalization layer, activation layer, and final regression layer.

We used normalization layer before the convolution to normalize the feature vector (H) over channels for all spatial locations. Normalization can be formulated as:

$$\mathbf{H}'_{ijk} = \frac{\mathbf{H}_{ijk}}{(k + a \sum_{k \in G(k)} \mathbf{H}_{ijk}^2)^\beta} \quad (6)$$

where β, a, K are normalization parameters and $G(k)$ is group of features over all channels.

Convolutional Layer consists of filters which perform 3D convolution on the normalized energy maps ($\mathbf{H}'$) obtained from the FFTEM stage. Convolution layers learn the patterns of input data and update the weights of the learned feature

TABLE I

COMPARISON OF CLASSIFICATION ACCURACY (KAPPA COEFFICIENTS) ON DATASET 2A BCI COMPETITION IV, WHERE 1ST TO 5TH ARE THE RESULTS FROM TOP FIVE POSITIONS IN BCI COMPETITION IV

Subjects	Proposed Method	[10]	[11]	DC[12]	OVR[12]	PW[12]	[13]	[14]	1st	2nd	3rd	4th	5th
1	0.70	0.88	0.62	0.71	0.68	0.78	0.70	0.74	0.68	0.69	0.38	0.46	0.41
2	0.45	0.22	0.28	0.37	0.42	0.40	0.45	0.43	0.42	0.34	0.18	0.25	0.17
3	0.76	0.88	0.66	0.66	0.75	0.75	0.71	0.80	0.75	0.71	0.48	0.65	0.39
4	0.58	0.39	0.33	0.41	0.48	0.52	0.96	0.57	0.48	0.44	0.33	0.31	0.25
5	0.72	0.53	0.14	0.40	0.40	0.41	0.60	0.31	0.40	0.16	0.07	0.12	0.06
6	0.51	0.33	0.25	0.26	0.27	0.19	0.43	0.38	0.27	0.21	0.14	0.07	0.16
7	0.63	0.38	0.41	0.73	0.77	0.80	0.55	0.76	0.77	0.66	0.29	0.00	0.34
8	0.56	0.85	0.60	0.58	0.75	0.74	0.61	0.75	0.75	0.73	0.49	0.46	0.45
9	0.60	0.81	0.66	0.50	0.61	0.54	0.57	0.80	0.61	0.69	0.44	0.42	0.37
mkv	0.61	0.58	0.44	0.52	0.57	0.57	0.62	0.61	0.57	0.52	0.31	0.30	0.29
stdev	0.101	0.268	0.19	0.165	0.183	0.212	0.192	0.196	0.183	0.229	0.153	0.213	0.134

over each iteration. 3D convolution can be formulated as:

$$\mathbf{Y}_{i'j'k'} = \sum_{ijk} \mathbf{W}_{ijk} \mathbf{H}'_{i+i',j+j',k} \quad (7)$$

where i, j, k are the dimensions of input features (energy maps), $\mathbf{W}$ is learned weight matrix, $\mathbf{H}'$ and $\mathbf{Y}$ are the input and output of convolution layer respectively.

Pooling layer is used to reduce the size of learned weights $\mathbf{Y}$. A window of size 2×2 is used with a stride of 2 and moved over the entire weight matrix $\mathbf{Y}$ with non-overlapping regions. Maximum value from each window is selected which has reduced the size of input weight matrix. Pooling function can be described mathematically using the following equation:

$$\mathbf{Z}_{ijk} = \max(\mathbf{Y}_{i'j'k'}) : i \leq i' < i+p, j \leq j' < j+p \quad (8)$$

where p is a constant used for padding (for odd number of rows or columns) and $\mathbf{Z}$ is the output with reduced dimensions.

Activation functions are used for non-linear mapping of weights $\mathbf{Z}$ on a predefined interval. Sigmoid, tanh, ReLU are some variants activation function. We tried different activation functions and found ReLU as the most optimal activation function in our case. After passing through the activation function, the matrix is converted into a vector $\mathbf{z}'$ using a dense layer.

For classification, we used softmax layer to compute softmax operator in a convolutional manner across all feature channels for all spatial locations. This layer consist of four neurons which computes the probability of similarity of input

query sample to all classes and the class corresponding to maximum probability is marked as the final class. Softmax operator can be described as:

$$s'_{ijk} = \frac{e^{\mathbf{z}_{ijk}}}{(\sum_{t=1}^D e^{\mathbf{z}_{ijt}})} \quad (9)$$

where D is the dimensions of input vector $\mathbf{z}$ and $\mathbf{s}'$ is the output of softmax layer. We used 10-fold cross-validation strategy for fair comparison.

E. Dataset Acquisition and Configuration

In this paper, we used BCI Competition Dataset 2a [7] (approved by Institutions Ethical Review Board) which consist of four class MI data: left hand, right hand, both feet, and tongue. This Dataset is collected from 9 subjects in 2 sessions and each session has 288 trials. The paradigm of BCI Competition Dataset 2a is shown in Fig. 2. Kappa coefficients are used as a benchmark to measure the accuracy as per competition requirements.

III. RESULTS AND DISCUSSION

We divided the Dataset into training (0.8) and testing (0.2) trials randomly. On testing Dataset, we achieved 0.61 mean kappa value. Multiple experiments have been conducted where different lengths and ranges of temporal segments and frequency bands are tried out while keeping other parameters constant. Which gave us a clear analysis of the impact of time segmentation on classification accuracy and optimal parameters are selected. For CNN classifier, different convolutional kernels and strides are used. Reasoning and motivation of all the selected parameters are summarized in Table II

Comparison of the proposed method with other techniques reported in the literature is given in Table I. In [10], researchers were able to achieve 0.58 mean kappa value by using two classifiers: first (KNN) for fusion of new information and second (SVM) for classification. Two simultaneous classifiers can increase the computational complexity. In [11], researchers introduced Separable Common Spatio-Spectral Pattern (SCSSP) for multi-class MI data and achieved 0.44 mean kappa value which is not a promising outcome as compared to other reported methods. In [12], researchers tried different classification methods i.e. Divide and Conquer

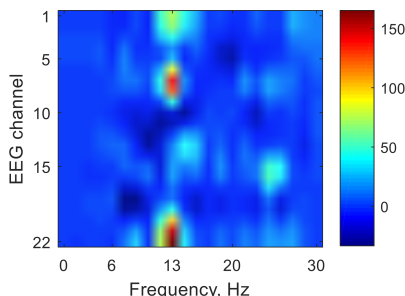


Fig. 3. Energy map of a single trial

TABLE II
EVALUATED DESIGN CHOICES

	Design aspect	Our choice	Reported variants	Reasoning/Motivation
Preprocessing	Temporal window	2.0-5.0 sec	subjective to user	to trim out the ques and segments where EEG signals are weak. Optimized after experimentation
	Frequency bands	$\mu(8 - 13), \beta(13 - 30)$	1-30 Hz	Reported in literature as most optimized bands. Also reduces the computational cost
	Filters	Buttherworth (5th order)	any type with any order	Response is smooth as compared to other orders and selected after multiple experiments
Features	Feature computation	CSP	CSP, FBCSP, LTCSP, Principle Component Analysis (PCA), Independent Component Analysis (ICA), Event-Related Synchronization/Desynchronization	Simple and efficient approach and less computational expensive as compared to ICA, PCA
	Feature selection	FFTEM	FFT, Band Power, Variance, Covariance, Energy, Correlation or any subjective to the user	Provides energy of bands corresponds to channels and results in spatial and spectral information
Classifier	Convolution	$7 \times 7 \times 16$	subjective to user	subjective to user
	Normalization	Batch normalization	single trial normalization, Batch normalization or without normalization	Batch normalization is better than a single trial. Reduces the variance of entire batch
	Activation	ReLU	Linear, tanh, eLU, ReLU, Leaky ReLU, Square	Reported best activation function for images
	Pooling	Max pooling	Average, max, mean	Returns maximum value of weights learned by the filters
	Classifier selection	CNN	Support Vector Machine (SVM), Linear Discriminant analysis (LDA), K-Nearest Neighbour (KNN)	Better than other classifiers for large and non-linear Datasets

(DC), Pair-Wise (PW) and One-Versus-Rest (OVR). they achieved maximum mean kappa value of 0.57 using OVR and PW classification methods. In [14], researchers achieved 0.61 mean kappa value by using Local Temporal Common Spatial Pattern (LTCSP) and LDA as feature extraction and classification technique respectively. In [13], researchers achieved 0.62 mkv by using Non-Negative Matrix Factorization (NNMF) for feature extraction and SVM as a classifier. In contrast, our proposed method utilized CNN followed by FFTEM and CSP and outperformed other techniques and in comparison to [14][13], our proposed model is robust to noise because of the less variance between the subjects and can be used as generalized method regardless of subject-specific limitations.

IV. CONCLUSION

We present a novel method for multi-class motor imagery classification. In order to extract features, we used CSP and FFTM to generate energy maps between frequency and channels. A Convolutional Neural Network (CNN) base model is proposed as a classifier for multi-class EEG data. In comparison to existing techniques, the proposed model can leverage the spatial and spectral information encoded in EEG data for classification of EEG signals, more accurately. We calculated FFTEM features for each trial against each subject after passing each trial from CSP to maximize co-variance among all classes using OVR mode. We extensively evaluate the proposed model on a publicly available benchmark Dataset. The proposed model has yielded mean kappa value of 0.61 and achieved the [14] best-reported results with lower computational complexity when compared with state-of-the-art methods. We conclude that our proposed method might be useful for real-time Motor Imagery classification.

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